Rivan Wong 黃家輝

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New Taipei City, Taiwan | Nationality: Indonesia

Summary

M.S. in Computer Science who builds software and the systems that verify it. Internship and industry co-op experience at Taiwan Mobile and Sunbird Software writing automated test systems against REST APIs and live services, and a master's thesis covering a method, its implementation, and six reimplemented baselines in PyTorch.

Education

National Taipei University of Technology (國立臺北科技大學) — *Taipei, Taiwan*
M.S. in Computer Science and Information Engineering — *2025 – 2026*

- GPA 4.0/4.0. Completed in one year through the university's selective Bachelor's–Master's program (學碩一貫學程); thesis advised by Shiang-Jiun Chen, Ph.D., in the Software Security Lab.

National Taipei University of Technology (國立臺北科技大學) — *Taipei, Taiwan*
B.S. in Computer Science and Information Engineering — *2021 – 2025*

- GPA 3.5/4.0. Member of the Software Security Lab.

Languages

Indonesian: Native | **English:** Fluent (TOEIC 965) | **Chinese (Mandarin):** Intermediate (TOCFL B2)

Technical Skills

Programming Languages: Python · C++ · Java · HTML · CSS
Testing & QA: Robot Framework · Selenium · Appium · PyTest · Parallel-safe test design · Test case design · UAT
Tools & Platforms: Git · GitHub · Jenkins · Jira · Notion · VS Code · REST APIs · Linux (Ubuntu) · Windows
Machine Learning & Research: PyTorch · Flower · Federated learning · Adversarial ML
Methodologies: Agile (Scrum/Kanban) · CI/CD · Acceptance-test-driven development · OOP · SDLC

Experience

Taiwan Mobile Co., Ltd. (台灣大哥大股份有限公司) — *Taipei, Taiwan*
Technical Project Management & QA Intern — *Jul 2025 – Jun 2026*

- Built the automated test suites for two other internal projects: Python with PyTest covering critical API endpoints, and Java automating a verification workflow previously done by hand.
- Owned pre-release verification for two internal AI products, one a VS Code extension that reviews code against the company's development standards in-editor, catching defects that compiled cleanly and read correctly in review: dropped formatting, truncated model output. Reported each to the engineers who built it and wrote the test reports the team worked from.
- Owned the release gate for the team building them: design agreed before code, explicit acceptance criteria, then hands-on sign-off, across roughly 40 changes and 300+ criteria over the two months they shipped in.
- Coordinated the six-person team that built both products as the technical link between the on-site engineers and the remote interns, managing sprint tasks in Jira so both sides worked from one view of state.

Sunbird Software Inc., Taiwan Branch (美商太陽鳥軟體股份有限公司台灣分公司) — *New Taipei, Taiwan*
Test Automation Intern, Industry-Academia Co-op via the Software Security Lab — *Jul 2023 – Dec 2024*

- Turned ticket-level requirement specifications into Selenium-driven end-to-end suites for an enterprise data-center web platform whose test bank runs to 730+ suites, composed from a shared library of thousands of reusable Robot Framework keywords, with state set up through a REST API facade rather than the UI.
- Wrote tests that stay safe in parallel, declaring by tag the shared subsystems each suite touches so hundreds run against one live server without colliding.
- Cleared the same merge gate every time: zero lint warnings, ten consecutive green runs, then five more under deliberate network throttling. Spent summer 2024 embedded full-time in the company's office.

Research & Projects

L-MAD: Trust-Aware Layer-Wise Median Absolute Deviation — Master's Thesis · 2025 – 2026

- Designed a two-stage defense for Byzantine-robust aggregation in federated edge learning, rejecting poisoned layers individually instead of discarding the whole client: holding 97.5% DDoS-class F_1 under a 30% poisoning attack: 25.7 points clear of the strongest baseline and 2.6× faster than the nearest layer-wise defense.
- Implemented the method and reimplemented all six competing defenses in one harness in PyTorch and Flower, then ran 21 experiment groups over 20 seeds with significance testing, closing with the five conditions under which the method fails.

Test Automation Best Practices: A Case Study Analysis — Capstone Project · 2023 – 2024

- Documented, gate by gate, how a commercial team takes a requirement ticket to a merged pull request, then analyzed four years of its output and found it had fallen by roughly two thirds from its peak with the process unchanged. Three students, advised by Shiang-Jiun Chen, Ph.D., from inside the Sunbird co-op.

Sky Scraper Escape — OOP Lab Course · C++, MFC · Mar – Jun 2024

- Fit seven levels and the menus onto a framework allowing exactly three game states, by mirroring its state interface onto a base class so every level became an interchangeable object behind one pointer.
- Wrote 76% of the 2,237 lines of C++ the two of us authored, including the base class, the player, and the menu system, integrated through 17 merged pull requests.